Gradually, as I neared the Wisconsin border, the pictures in my mind became clear. The wormhole would *not* be destroyed. Geroch and Wald had overlooked a crucial fact: Every time the beam of radiation passes through the wormhole, the wormhole *defocuses* it in the manner of Box 14.1 above. After the defocusing, the beam emerges from the mouth on Earth and spreads out over a wide swath of space, so that only a tiny fraction of it can get caught by the mouth on the spacecraft and transported through the wormhole back to Earth to "pile up" on itself (Figure 14.8b).

I could do the sum visually in my head, as I drove. By adding up all the radiation from all the trips through the wormhole (a tinier and tinier amount after each defocusing trip), I computed that the final beam would be weak; far too weak to destroy the wormhole.

My calculation turned out to be right; but, as I shall explain later, I should have been more cautious. This brush with wormhole destruction should have warned me that unexpected dangers await any maker of time machines.

When graduate students reach the final year of their research, they often give me great pleasure. They produce major insights on their own; they argue with me and win; they teach me unexpected things. Such was the case with Morris and Yurtsever as we gradually moved toward finalizing our manuscript for *Physical Review Letters*. Large portions of the manuscript's technical details and technical ideas were theirs.

As our work neared completion, I oscillated between worrying about tarnishing Morris's and Yurtsever's budding scientific reputations with a label of "crazy science fiction physicists" and waxing enthusiastic about the things we had learned and about our realization that Sagan-type questions can be powerful in physics research. At the last minute, as we finalized the paper, I suppressed my caution (which Morris and Yurtsever seemed not to share), and agreed with them to give our paper the title "Wormholes, Time Machines, and the Weak Energy Condition" ("weak energy condition" being the technical term associated with "exotic matter").

Despite the "time machines" in the title, our paper was accepted for publication without question. The two anonymous referees seemed to be sympathetic; I heaved a sigh of relief.

With the publication date nearing, caution took hold of me again; I asked the staff of the Caltech Public Relations Office to avoid and, indeed,